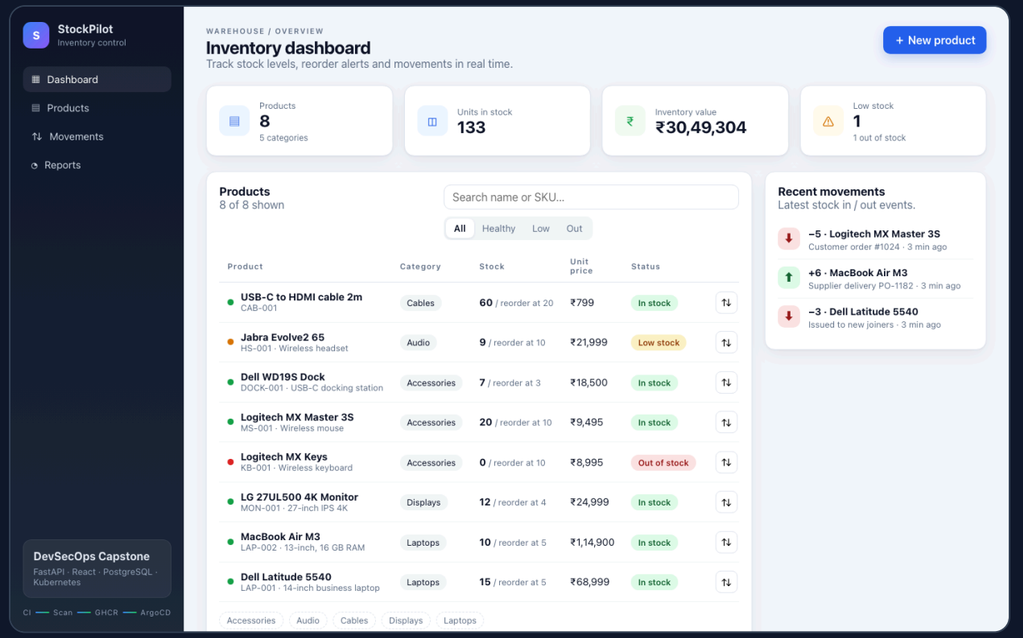
InvenPilot — Inventory Management

DevSecOps Final Capstone · Gaurav Kumar

github.com/Gaurakumar-s/invenpilot-devsecops

FastAPI · React · PostgreSQL · Docker · GitHub Actions · Trivy · GHCR · Terraform · AWS EKS · Helm · ArgoCD · Prometheus · Grafana



InvenPilot dashboard running on AWS EKS

Project Overview

PROBLEM

Small warehouses track stock in spreadsheets: no single source of truth and no reorder alerts, so items run out unnoticed.

SOLUTION — INVENPILOT

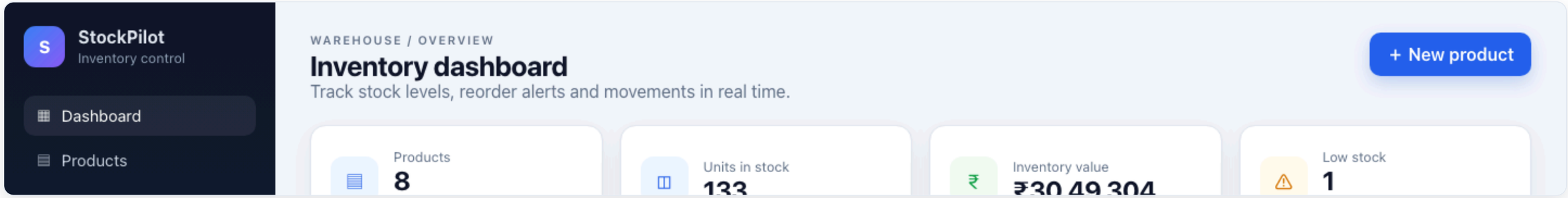
A web app that keeps a live product catalogue, records every stock movement, and flags what needs reordering.

FEATURES

- Products with SKU, category, quantity, reorder level, unit price
- Stock IN / OUT movements ledger
- Low-stock and out-of-stock alerts
- KPI dashboard: products, units, inventory value, low stock
- Responsive React UI with search and filters

REST API — 9 ENDPOINTS

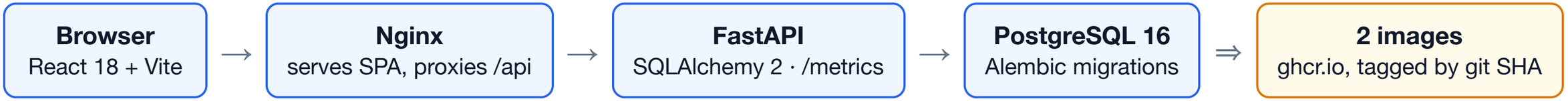
- GET / POST / PUT / DELETE `/api/products`
- GET `/api/products/stats` · GET `/api/products/low-stock`
- POST `/api/products/{id}/movements` · GET `/api/movements`
- Ops: `/health` · `/ready` · `/metrics` · `/docs`



Tech Stack

Application	DevOps / Platform
React 18 + Vite (frontend SPA)	Docker — multi-stage builds, non-root containers
FastAPI (Python REST API)	Docker Compose (local stack)
SQLAlchemy 2 + Alembic (2 migrations)	GitHub Actions (CI/CD) · Trivy (image scanning) · GHCR
PostgreSQL 16	Terraform — AWS VPC + EKS
pytest (20 tests) · ruff (lint)	Helm · ArgoCD · ingress-nginx · HPA
	Prometheus + Grafana (kube-prometheus-stack)

RUNTIME REQUEST PATH



Architecture

DELIVERY PATH

Developer
git push

→

GitHub
source + Helm values

→

GitHub Actions
test · build · Trivy · push · bump tag

→

GHCR
images by git SHA

→

ArgoCD
watches repo, auto-sync

AWS EKS — TERRAFORM-PROVISIONED VPC (2 PUBLIC + 2 PRIVATE SUBNETS, NAT) · NAMESPACE INVENPILOT

User / browser
ELB → Ingress

→

Ingress (nginx)
/ → frontend
/api → backend

→

frontend
Nginx · 2 replicas

backend
FastAPI · 2 replicas
HPA 2–6 (CPU 60%)

→

PostgreSQL
PVC via EBS CSI

Prometheus
ServiceMonitor scrapes backend /metrics

→

Grafana
InvenPilot dashboard

namespace monitoring

CI/CD Pipeline — GitHub Actions

Triggers: push to main · pull request · manual (workflow_dispatch)

1 · backend-test
ruff lint + pytest (20 tests)
red tests fail the build

→

2 · frontend-build
npm ci + vite build

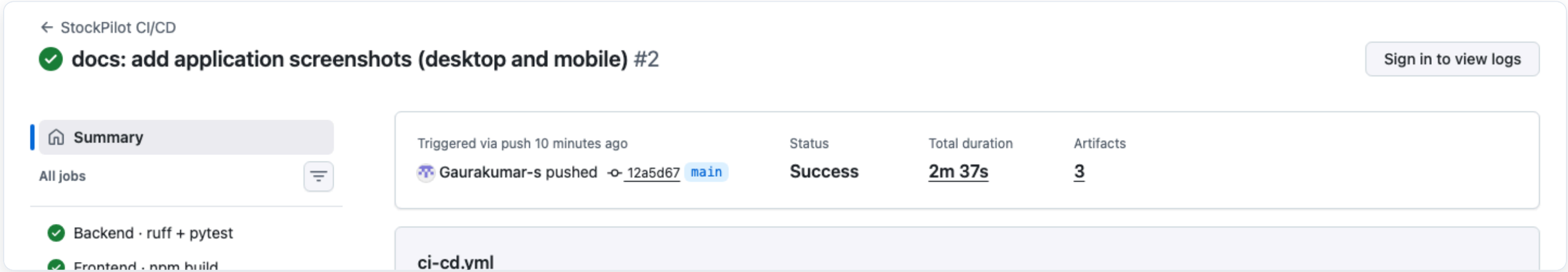
→

3 · build-scan-push
docker build both images
Trivy: HIGH/CRITICAL → exit 1
push to GHCR tagged :<git-sha>

→

4 · update-gitops
write SHA into helm values
commit → ArgoCD auto-syncs

- Jobs 1 and 2 run in parallel; job 3 needs both green; job 4 needs job 3 and runs only on main.
- Images: `ghcr.io/gaurakumar-s/invenpilot-backend:<git-sha>` and `…-frontend:<git-sha>` — never `latest`, so every pod traces back to one commit.
- Nothing is pushed unless tests and scans pass. The pipeline never runs kubectl: it changes Git, ArgoCD changes the cluster.



DevSecOps — Security Controls

TRIVY IMAGE SCANNING IN CI

- Scans OS packages + Python/npm libraries of both images
- Pipeline fails on HIGH / CRITICAL (ignore-unfixed)
- **Finding during build:** starlette 0.46 DoS (CVE-2025-62727) and old Alpine 3.21 packages (musl, zlib, libxml2)
- **Fix:** upgraded to starlette 1.8 / FastAPI 0.143 and nginx-unprivileged on Alpine 3.24

Scan status: 0 HIGH / 0 CRITICAL

OTHER CONTROLS

- Non-root containers: backend uid 10001, frontend uid 101
- securityContext drops ALL capabilities, no privilege escalation
- No secrets in git: .gitignore, terraform.tfvars.example, credentials only in the AWS CLI profile
- Least-privilege GITHUB_TOKEN (packages: write only in the push job)
- Immutable SHA tags; GitOps means no manual cluster access in CI

```
Report Summary
| ghcr.io/gaurakumar-s/stockpilot-backend:ab18a8dd424ce39edaff791952803e42e624fe75 |   debian   |       0       |   -   |
Legend:
- '0': Clean (no security findings detected)
  image-ref: ghcr.io/gaurakumar-s/stockpilot-frontend:ab18a8dd424ce39edaff791952803e42e624fe75
  INPUT_IMAGE_REF: ghcr.io/gaurakumar-s/stockpilot-frontend:ab18a8dd424ce39edaff791952803e42e624fe75
Running Trivy with options: trivy image ghcr.io/gaurakumar-s/stockpilot-frontend:ab18a8dd424ce39edaff791952803e42e624fe75
Report Summary
| ghcr.io/gaurakumar-s/stockpilot-frontend:ab18a8dd424ce39edaff791952803e42e624fe- | alpine |       0       |   -   |
```

Trivy output from the GitHub Actions run: both images clean (0 findings) before the push to GHCR

Infrastructure — Terraform on AWS

NETWORK (AP-SOUTHEAST-2)

- VPC 10.20.0.0/16 across 2 Availability Zones
- 2 public subnets — load balancer (ELB)
- 2 private subnets — EKS worker nodes
- Single NAT gateway for node egress

WORKFLOW

- terraform init → validate → plan → apply → destroy
- terraform.tfvars.example committed; no credentials in git
- ≈ $0.35 / hour — cluster destroyed after the demo

EKS CLUSTER

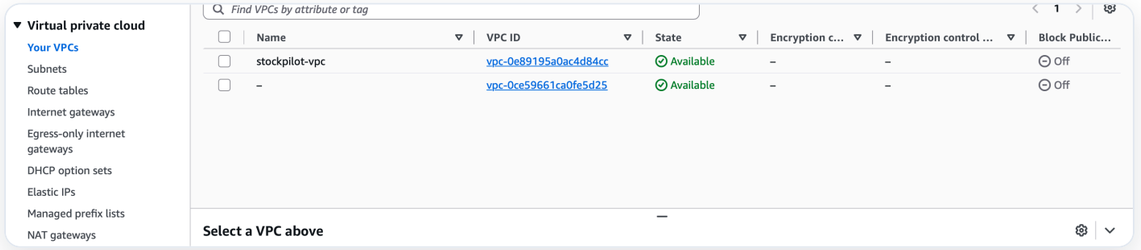
- EKS 1.36, managed node group
- m7i-flex.large × 2 (free-tier eligible, 2 vCPU / 8 GiB; min 2, max 2)
- Add-ons: coredns, kube-proxy, vpc-cni, aws-ebs-csi-driver (EBS policy on the node role) — backs the Postgres PVC

VPC 10.20.0.0/16

public ×2
ELB + NAT

private ×2
EKS nodes

AZ-a and AZ-b each hold one public and one private subnet. Region is fixed by the account's organization policy.



AWS console: invenpilot-vpc in ap-southeast-2

Kubernetes · Helm · ArgoCD

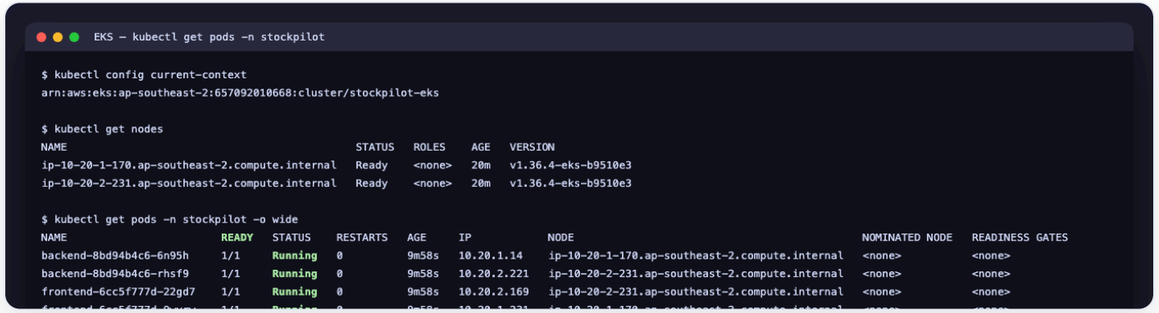
HELM CHART — HELM/INVENPILOT

- Deployments: backend ×2, frontend ×2, postgres ×1 + PVC
- ClusterIP Services; Secret for DB credentials
- Ingress (nginx class): / → frontend, /api → backend
- HPA on backend CPU 60 %: 2 → 6 replicas
- ServiceMonitor for Prometheus; liveness + readiness probes
- values-minikube.yaml / values-eks.yaml

ARGOCD — GITOPS

- Application points at the chart in the repo (main)
- Automated sync + prune + selfHeal
- Git is the source of truth: drift is reverted, deletions pruned

```
$ kubectl get pods,svc,ingress -n invenpilot
$ helm list -n invenpilot
$ kubectl get hpa -n invenpilot
$ argocd app get invenpilot
```





Observability — Prometheus + Grafana

METRICS PIPELINE

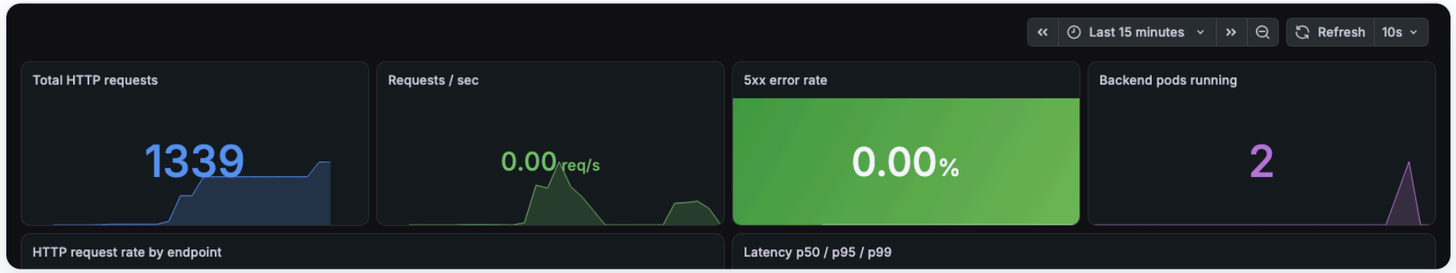
- Backend exposes /metrics via prometheus-fastapi-instrumentator
- http_requests_total, http_request_duration_seconds
- kube-prometheus-stack in namespace monitoring
- ServiceMonitor (release=kube-prometheus-stack) → Targets page: backend UP
- load-test.sh generates traffic for the panels

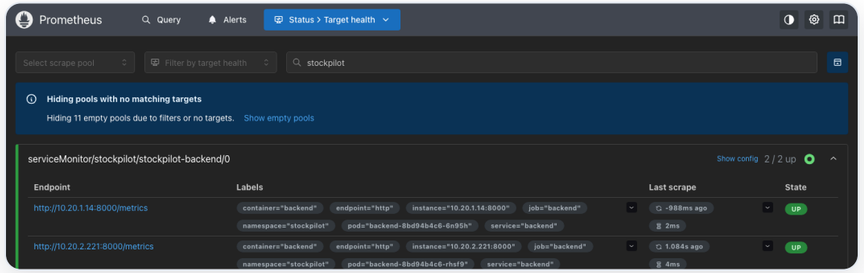
backend /metrics → **ServiceMonitor** → **Prometheus** → **Grafana**

GRAFANA — "INVENPILOT — APPLICATION METRICS"

Requests / sec	p50 / p95 / p99 latency
5xx error rate	Responses by status code
Backend CPU (HPA input)	Pods running

Six panels on one dashboard: traffic, latency, errors, and the CPU signal that drives the HPA.





Key Learnings & Demo Agenda

KEY LEARNINGS

- Quality gates before promotion: tests → scan → registry
- Immutable SHA tags give end-to-end traceability
- GitOps removes manual kubectl deploys
- PVCs on EKS need the EBS CSI driver
- Cloud accounts have real constraints: region policy, free-tier instance types, quotas
- Always destroy cloud infrastructure when done

LIVE DEMO AGENDA

1. Codebase tour
2. Push a commit, watch GitHub Actions
3. GHCR: new SHA-tagged images
4. terraform plan + AWS console
5. ArgoCD: synced application
6. kubectl pods / services / ingress, app via load balancer
7. Prometheus targets
8. Grafana dashboard

Thank you · github.com/Gaurakumar-s/invenpilot-devsecops